Correlation and Multicollinearity

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Table of contents

1 Why This File? **1**

2 Data **2**

3 Correlation Matrix **2**
3.1 What it shows 2
3.2 All variables 3
3.3 Only selected variables 3

4 pairs.panels(): All Relationships at Once **3**
4.1 How to read the three parts 4

5 VIF (Variance Inflation Factor) **5**
5.1 Fit the model and get VIF 5
5.2 Meaning of VIF values 6
5.3 Where does VIF come from? 6

6 What Next if Multicollinearity is High? **6**

7 Key Takeaways **6**

1 Why This File?

In file 01 we saw a model with a very high R^2 but **no significant individual predictor**. The usual reason: the predictors **overlap** (they carry the same information).

This file shows how to **detect** that problem:

1. **Correlation matrix**: how strongly are variables related to each other?
2. **pairs.panels()**: see all relationships at once.

3. **VIF**: how redundant is each predictor?

How to fix it is in file 03.

```
library(psych)
library(car)
library(ggplot2)
library(knitr)
```

2 Data

```
data <- read.csv("../data/raw/high_correlation_agriculture.csv")
head(data)
```

```
  yield nitrogen rainfall temperature
1 5.561    84.89   580.76       25.65
2 5.374    73.06   570.03       24.69
3 6.096    89.08   609.58       23.65
4 6.222    89.55   645.73       24.70
5 5.911    88.63   602.24       23.36
6 5.293    77.18   609.65       25.23
```

3 Correlation Matrix

3.1 What it shows

The correlation matrix shows **how strongly variables are related to each other**. Values range from -1 to +1:

Value	Meaning
+1	perfect positive correlation: perfect increase-increase relation
0	no correlation: no clear relationship
-1	perfect negative correlation: one increases, the other decreases

Intuition: if two variables rise and fall together, r is near +1. If one rises while the other falls, r is near -1. If they ignore each other, r is near 0.

3.2 All variables

```
cor_mat <- cor(data)
kable(round(cor_mat, 3))
```

	yield	nitrogen	rainfall	temperature
yield	1.000	0.976	0.961	0.048
nitrogen	0.976	1.000	0.968	0.067
rainfall	0.961	0.968	1.000	0.095
temperature	0.048	0.067	0.095	1.000

3.3 Only selected variables

```
cor(data[, c("yield", "rainfall")])
```

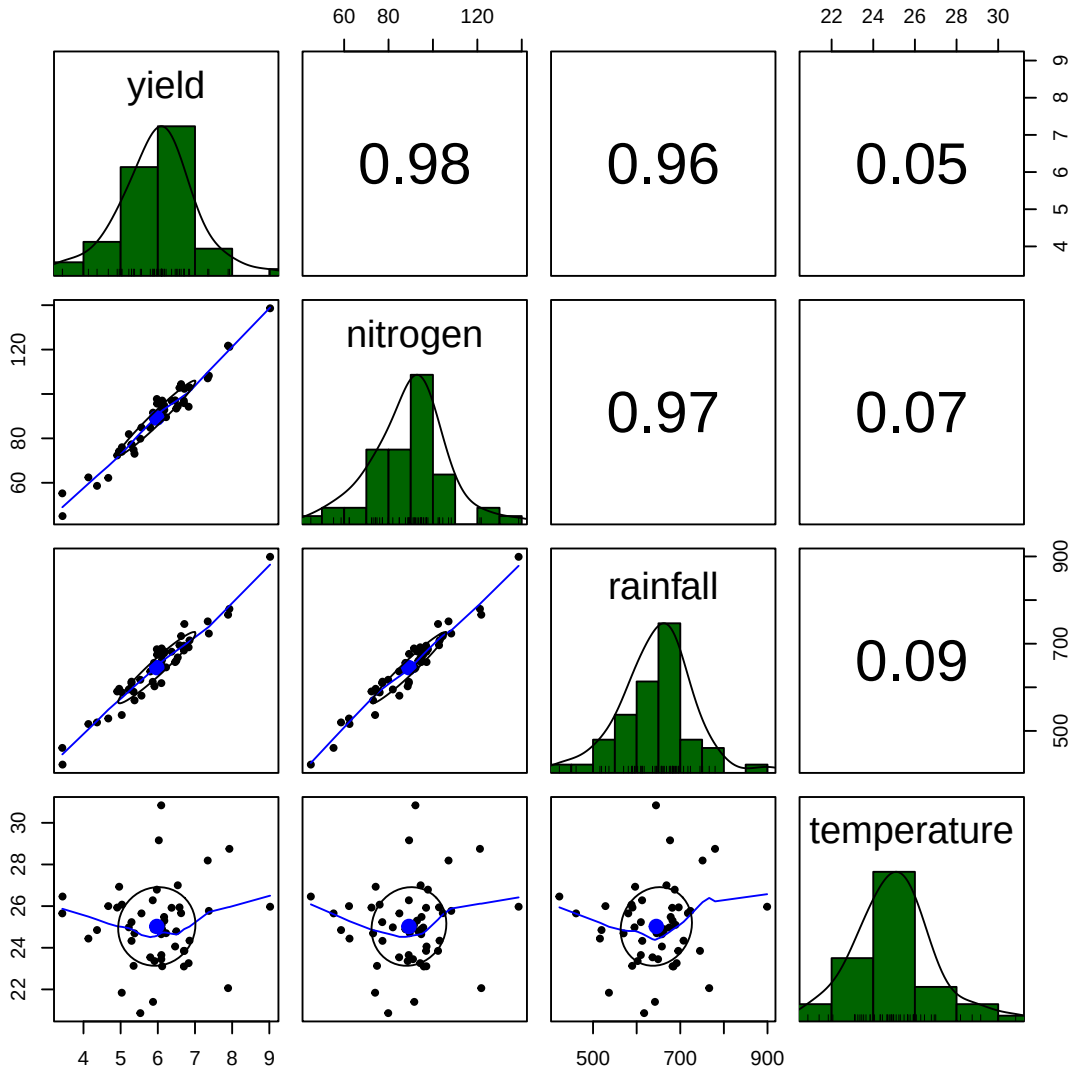
```
             yield  rainfall
yield    1.0000000 0.9611725
rainfall 0.9611725 1.0000000
```

Important: a high correlation between a predictor and **yield** is good (the predictor is useful). A high correlation between two **predictors** (for example nitrogen and rainfall) is a warning sign: they may be redundant.

4 pairs.panels(): All Relationships at Once

`pairs.panels()` shows **all variable relationships at once**. `ggplot` shows **one relationship in detail**.

```
pairs.panels(data,
             method = "pearson",
             hist.col = "darkgreen",
             density = TRUE,
             ellipses = TRUE)
```



4.1 How to read the three parts

Upper triangle: correlation coefficients

- Shows the **strength and direction** of the relationship between two variables.
- Range: +1 perfect positive, 0 none, -1 perfect negative.

Diagonal: histograms (with density lines if `density = TRUE`)

- Shows the **distribution** of each variable.
- Use it to check:
  - **Normality** (bell-shaped)
  - **Skewness** (leaning left or right)
  - **Outliers** (isolated bars)

Lower triangle: scatter plots (with ellipses if `ellipses = TRUE`)

- Visualizes the **trend** between pairs of variables.
- Line-shaped or elliptical = strong linear relation.
- Wide or spread = weak or no correlation.
- Outliers appear as distant points.

Ellipses:

- **Thinner** = stronger linear relationship.
- **Round** = weak or no relationship.
- The **direction** shows a positive or negative trend.

5 VIF (Variance Inflation Factor)

- VIF measures **how much one predictor can be predicted by the other predictors**.
- The higher the VIF, the more **redundant (overlapping) information** that predictor has.

Think of two people giving you the same news. The second person adds no new information. If your predictors are like that, the model cannot tell **whose effect is whose**, so the coefficients become unstable and look "not significant".

5.1 Fit the model and get VIF

```
model <- lm(yield ~ nitrogen + rainfall, data = data)
vif(model)
```

```
nitrogen rainfall
16.11817 16.11817
```

5.2 Meaning of VIF values

VIF value	Interpretation
1	No correlation with other variables (ideal)
1 to 5	Moderate correlation, usually okay
> 5	High correlation, possible problem
> 10	Serious multicollinearity, bad for regression

5.3 Where does VIF come from?

VIF for a predictor = 1 / (1 - R^2), where R^2 comes from predicting **that predictor** using the **other predictors**.

The manual VIF matches `vif()`. So: **if rainfall can almost fully predict nitrogen (R^2 close to 1), VIF explodes**.

6 What Next if Multicollinearity is High?

The next step is **managing** it (file 03):

1. Collect more data.
2. Remove one variable, or combine variables into one.
3. PCA (Principal Component Analysis).
4. Ridge regression.

7 Key Takeaways

1. **Correlation matrix** (`cor()`): strength and direction between variables, from -1 to +1.
2. **pairs.panels()**: upper triangle = coefficients, diagonal = histograms (normality, skewness, outliers), lower triangle = scatter plots and ellipses (thin ellipse = strong relation).
3. **Predictor-yield** correlation high = useful. **Predictor-predictor** correlation high = redundancy warning.
4. **VIF**: 1 is ideal, 1 to 5 usually okay, above 5 a possible problem, above 10 serious.